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Opinion: Alberta's AI boom poses hidden health, environmental risks

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Planned data centres pose hidden health and environmental risks, says an Edmonton physician. Getty Images/iStockphoto

As a doctor, protecting human health sits at the heart of what I do. This commitment extends beyond hospital walls to the air we breathe, the water we drink, and the communities we build. Today, I'm concerned about a growing threat to health: the rush to build gas-powered AI data centres across Alberta.

Alberta's government has embraced these projects as economic drivers, with proposals like Kevin O'Leary's \$70-billion "[Wonder Valley](#)" project promising jobs and investment. In her [recent throne speech](#), Premier Danielle Smith described "using our almost inexhaustible supplies of natural gas to fuel the massive amounts of power required for AI data centres."

Missing from these announcements is any discussion of the costs these facilities, and continued reliance on fossil fuels, will impose

on communities and our province.



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Gas-powered data centres are not benign economic engines. They are industrial facilities that release harmful pollutants into our air and water, starting with the gas turbines that power them. I'm particularly concerned about three categories of emissions that pose risks to public health.

First, [nitrogen oxides from gas turbines](#) worsen respiratory diseases like asthma and chronic obstructive pulmonary disease (COPD), which already send thousands to Alberta emergency rooms each year.

Second, [turbines emit fine particulate matter](#)— PM2.5, similar to wildfire smoke — which can enter the bloodstream and [contribute to heart attacks and strokes](#). I see patients with these conditions daily in the emergency department, where I witness first-hand the suffering our communities will face if air pollution worsens.

Third, these facilities release volatile organic compounds (VOCs), including formaldehyde, benzene, and toluene. Communities exposed to higher concentrations of these VOCs face [increased risks of cancer](#) and other chronic illnesses. We've all experienced smoke-filled summers in recent years — why would we deliberately add more harmful pollution to our air?

Air pollution from powering these facilities isn't the only concern. [Recent reporting](#) has highlighted another threat: PFAS, “forever chemicals,” used in data-centre cooling systems and electronic components. These substances persist in the environment and are linked to cancer, birth defects, thyroid disease, and weakened immune systems.

[Evidence from the U.S.](#) suggests PFAS from data centres may leak, accumulate, and travel through air and water, risking long-term contamination. If Alberta follows a similar build-out without safeguards, we risk releasing these toxins into our environment and water systems.

This is also an environmental justice issue. These projects are being placed far from urban centres where decisions are made,

leaving rural and Indigenous communities to bear the health burden. The Sturgeon Lake Cree Nation, for instance, reported [“zero consultation”](#) on the Wonder Valley proposal on their traditional territory.

In Memphis, [Elon Musk’s xAI](#) built gas-powered data centres near historically Black communities without proper permits or pollution controls, prompting legal action by the NAACP for environmental racism. We need to learn from these examples and prevent similar injustice from happening in Alberta.

Proponents often claim economic necessity, but this ignores both health costs and shifting energy economics. Analyst [Amory Lovins warns](#) that Alberta’s gas-centric strategy for data centres is likely to fail, as renewables now out-compete fossil generation. By doubling down on gas, Alberta risks “killing the data-centre industry they’re hoping to create.”

As physicians, we’re already seeing the impacts of climate change in our practices, from heat-related illnesses to respiratory problems from wildfire smoke. Gas-powered, PFAS-emitting data centres would compound these problems while locking us into decades more fossil fuel dependence.

Alberta deserves better. We need comprehensive environmental impact assessments for all proposed data centres, including full disclosure of chemical use and meaningful consultation with Indigenous and other affected communities. Any new facilities should be powered by renewables and avoid the use of PFAS, creating jobs while protecting health. [Other jurisdictions are already moving in this direction](#), recognizing that clean energy is healthier and more economical.

Last year, Alberta celebrated the end of coal-fired electricity in our province, a public-health victory that many of us fought hard to achieve. Let's not replace one source of harmful pollution with another. Our health, our environment, and our future depend on making wiser choices today.

Julia Sawatzky is an Edmonton-based resident doctor training in emergency medicine. She is a board member of the Canadian Association of Physicians for the Environment (CAPE) and co-chair of CAPE Alberta.

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